

PROBLEM STATEMENT #48 | Developer Tools & IT Operations

Incident Escalation & On-Call Rotation Engine

1. Problem Context & Overview

A DevOps on-call management system organizing team shift rotations, routing inbound monitoring alerts through multi-tiered phone/SMS escalation ladders, and tracking incident post-mortems.

Target Stakeholders / Actors: *On-Call Engineer, Incident Commander*

2. Sample Functional Requirement (FR) Guideline

Sample Requirement: FR-001 [Priority: High]

Description: The system shall ingest incident alerts, identify the active on-call engineer from the rotation schedule, and escalate to secondary engineer if unacknowledged within 5 minutes.

Sample Acceptance Criteria: Pass: Escalation step triggered upon timeout; Fail: Unacknowledged P1 incident drops without escalation.

3. Sample Non-Functional Requirement (NFR) Guideline

Sample Requirement: NFR-001 [Type: Performance & Security]

Description: Alert dispatch via webhook, SMS, and email must initiate within 3 seconds of alert ingestion.

Sample Acceptance Criteria: Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.

Deliverables to Submit: [All through a GitHub Repo in your account]

- **Complete Requirements Table:**

- >> Exactly 5 Functional Requirements (FR-001 to FR-005)
- >> 2 Non-Functional Requirements (NFR-001 & NFR-002) with ID, Type, Description, Priority, Acceptance Criteria, and Rationale.

- **UML Use-Case Diagram:**

- >> Model all actors, primary use cases, and include at least one «include» and one «extend» relationship.

- **Use-Case Flow Specification:**

- >> 1-page document for one core use case detailing Preconditions, Postconditions, Main Success Scenario
- >> One Alternate Flow.